

SYDNEY GRAMMAR SCHOOL



2016 Annual Examination

FORM V

MATHEMATICS EXTENSION 1

Monday 5th September 2016

General Instructions

- Writing time — 2 hours
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 100 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and Master on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your name, class and Master on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

5A: DNW
5E: WJM5B: PKH
5F: GMC5C: LRP
5G: NL5D: FMW
5H: SO**Checklist**

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 141 boys

Examiner
FMW

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

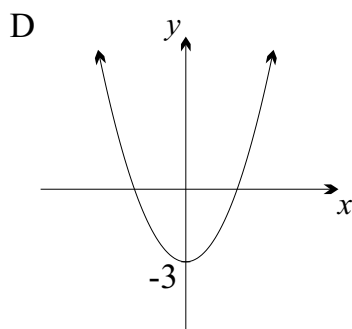
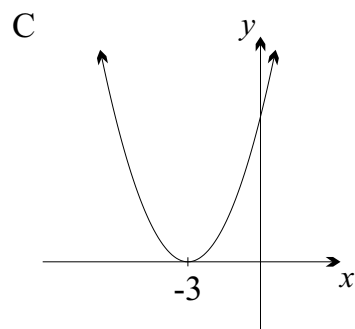
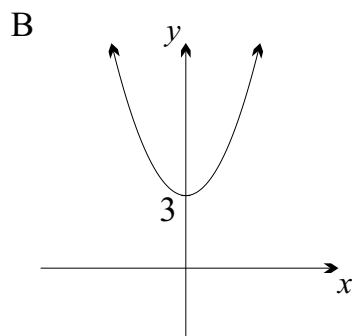
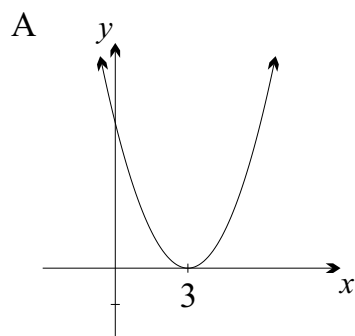
QUESTION ONE

What is the value of $20e^{-2}$, correct to 3 significant figures?

- (A) 2.71
- (B) 2.70
- (C) 2.707
- (D) 2.706

QUESTION TWO

Which graph below best represents $y = (x + 3)^2$?



Examination continues next page ...

QUESTION THREE

What are the solutions of $3x^2 - 6x + 2 = 0$?

(A) $x = \frac{3 \pm \sqrt{15}}{3}$

(B) $x = \frac{-3 \pm \sqrt{3}}{3}$

(C) $x = 1 \pm 2\sqrt{3}$

(D) $x = \frac{3 \pm \sqrt{3}}{3}$

QUESTION FOUR

Given $y = \frac{x^2}{x-1}$, then $\frac{dy}{dx}$ is equal to:

(A) $2x$

(B) $\frac{x-2}{x}$

(C) $\frac{x(2-x)}{(x-1)^2}$

(D) $\frac{x(x-2)}{(x-1)^2}$

QUESTION FIVE

In relation to the function $y = |2x - 3|$, which of the following statements is NOT true?

(A) The domain of the function is all real x .

(B) The range of the function is $y \geq 0$.

(C) The function is continuous for all values of x .

(D) The function is differentiable for all values of x .

Examination continues overleaf ...

QUESTION SIX

What is the solution of $7^x = 2$?

(A) $x = \log_2 7$

(B) $x = \log_{10} \frac{2}{7}$

(C) $x = \frac{\log_{10} 2}{\log_{10} 7}$

(D) $x = \frac{\log_e 7}{\log_e 2}$

QUESTION SEVEN

Which of the following is a primitive of $\frac{1}{x\sqrt{x}}$?

(A) $\log_e x\sqrt{x}$

(B) $-\frac{2}{\sqrt{x}}$

(C) $-\frac{3}{2}x^{-\frac{5}{2}}$

(D) $-\frac{2}{5}x^{-\frac{5}{2}}$

QUESTION EIGHT

How many solutions does the equation $\cos 2\theta = 1$ have in the domain $0^\circ \leq \theta \leq 360^\circ$?

(A) 1

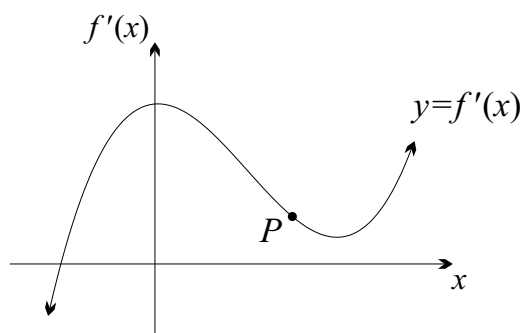
(B) 2

(C) 3

(D) 4

Examination continues next page ...

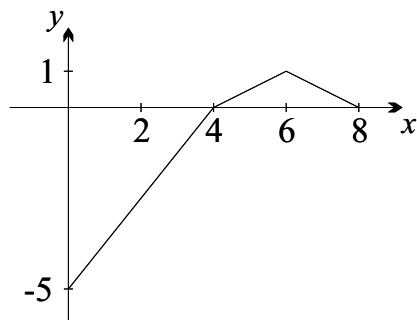
QUESTION NINE



The GRADIENT FUNCTION $y = f'(x)$ of a curve is sketched above. Which of the following statements is true at the point P ?

- (A) $f'(x) < 0$ and $f''(x) > 0$
- (B) $f'(x) > 0$ and $f''(x) < 0$
- (C) $f'(x) > 0$ and $f''(x) > 0$
- (D) $f'(x) < 0$ and $f''(x) < 0$

QUESTION TEN



The function $y = f(x)$ is sketched above. Which of the following integrals yields the greatest value?

- (A) $\int_0^2 f(x) dx$
- (B) $\int_0^4 f(x) dx$
- (C) $\int_0^6 f(x) dx$
- (D) $\int_0^8 f(x) dx$

_____ End of Section I _____

Examination continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet. **Marks**

(a) Expand and simplify $(2\sqrt{3} - 5)(2\sqrt{3} + 5)$. **1**

(b) Write down the equation of the line which is parallel to $3x - y + 7 = 0$ and has a y -intercept of -5 . **1**

(c) Write down a quadratic equation with roots 5 and -2 . **1**

(d) Simplify:

(i) $\log_e e^3$ **1**

(ii) $e^{\log_e 5}$ **1**

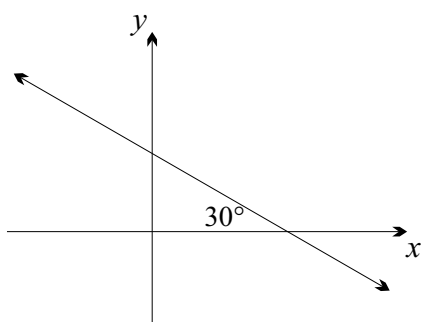
(e) Write down the equations of the vertical asymptotes of the graph of the function $f(x) = \frac{1}{x^2 - 16}$. **1**

(f) Solve $\cos \theta = \frac{1}{\sqrt{2}}$, for $0^\circ \leq \theta \leq 360^\circ$. **1**

(g) Find the gradient of the interval joining the points $(1, 2)$ and $(-2, -4)$. **1**

(h) Find the sum of the first 100 terms of the arithmetic sequence $12, 6, 0, \dots$. **2**

(i) **1**



Write down the gradient of the line sketched above.

(j) Find the gradient of the tangent to the curve $y = e^{2x+1}$ at the point on the curve where $x = -1$. **2**

(k) (i) Sketch $y = e^x$. **1**

(ii) Evaluate $\int_0^1 e^x dx$. **1**

Examination continues next page ...

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**

(a) Differentiate:

(i) $y = \frac{3}{x}$

1

(ii) $y = \ln(5x - 1)$

1

(iii) $y = xe^x$

1

(b) Find:

(i) $\int (2x - 5)^4 dx$

1

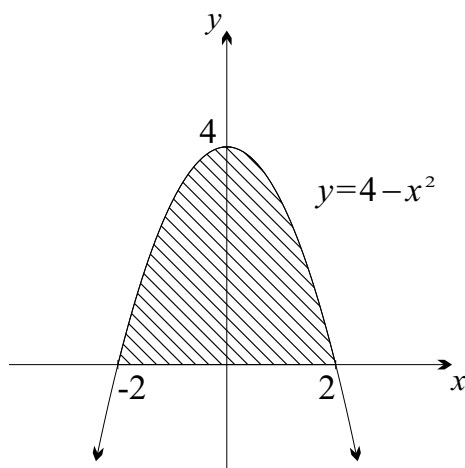
(ii) $\int \frac{2}{x-1} dx$

1

(iii) $\int \frac{x-1}{x^2} dx$

2(c) Find the equation of the curve passing through the point $(-1, 4)$ with gradient function $\frac{dy}{dx} = 3x^2 + 4x - 5$.**3**

(d)

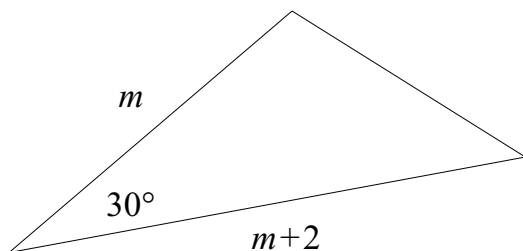
2Find the area bounded by the curve $y = 4 - x^2$ and the x -axis, as shaded above.(e) Find the values of k for which the quadratic equation $x^2 + kx + 36 = 0$ has two distinct, real roots.**3**

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

(a)

3



The area of the triangle above is 12 square units. Find the value of m .

- (b) (i) Copy and complete the following table of values for $y = \log_e x$, giving your answers correct to 2 decimal places where necessary.

1

x	1	2	3
y			

- (ii) Use the trapezoidal rule with all the values from your table to find an approximation for $\int_1^3 \log_e x \, dx$.

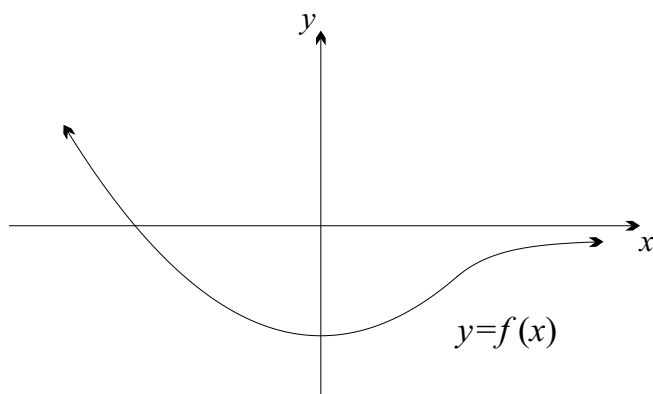
2

- (iii) Explain, with the use of a diagram, whether the approximation in (ii) is less than or greater than the exact value.

1

(c)

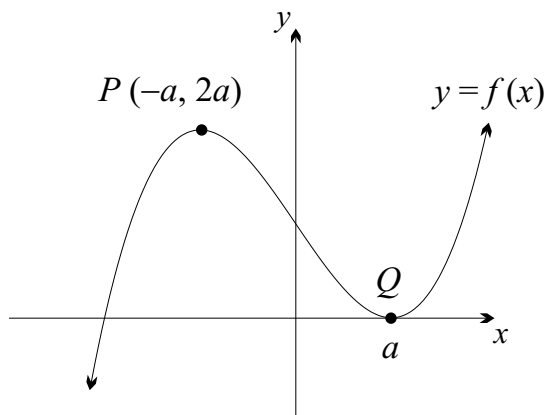
2



The diagram above shows the curve $y = f(x)$. Sketch a possible graph of $y = f'(x)$.

- (d) (i) Given the function $g(x) = 2x^3 - 3x^2 - 12x + 4$, find $g'(x)$ and $g''(x)$. 1
- (ii) For what values of x is $y = g(x)$ decreasing? 2
- (iii) For what values of x is $y = g(x)$ concave up? 1

(e)

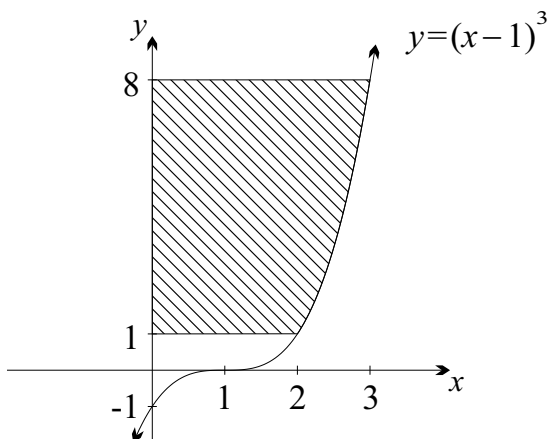


The graph of $y = f(x)$ is sketched above with points P and Q as marked.

- (i) Write down the coordinates of point P under the transformation $y = -2f(x)$. 1
- (ii) Write down the coordinates of point Q under the transformation $y = f(-x)$. 1

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks****3**

(a)



The diagram above shows the curve $y = (x - 1)^3$. Find the shaded area.

(b) Let α and β be the roots of the quadratic equation $2x^2 - 4x + 7 = 0$. Find:

(i) $\alpha + \beta$

1

(ii) $\alpha\beta$

1

(iii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

2

(c) Show algebraically that the curve $y = (2 - x)^3$ has a point of inflexion at $(2, 0)$.

2

(d) Find the coordinates of any stationary points on the curve $y = x^2 + \frac{16}{x}$ and determine their nature.

3

(e) Use a suitable substitution to solve the equation $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 6 = 0$.

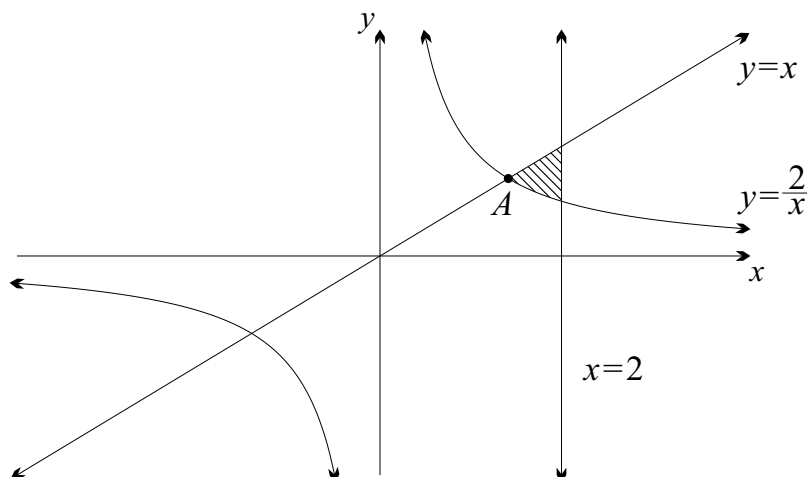
3

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) Factorise completely
- $3b^3 - 24$
- .

2

- (b)



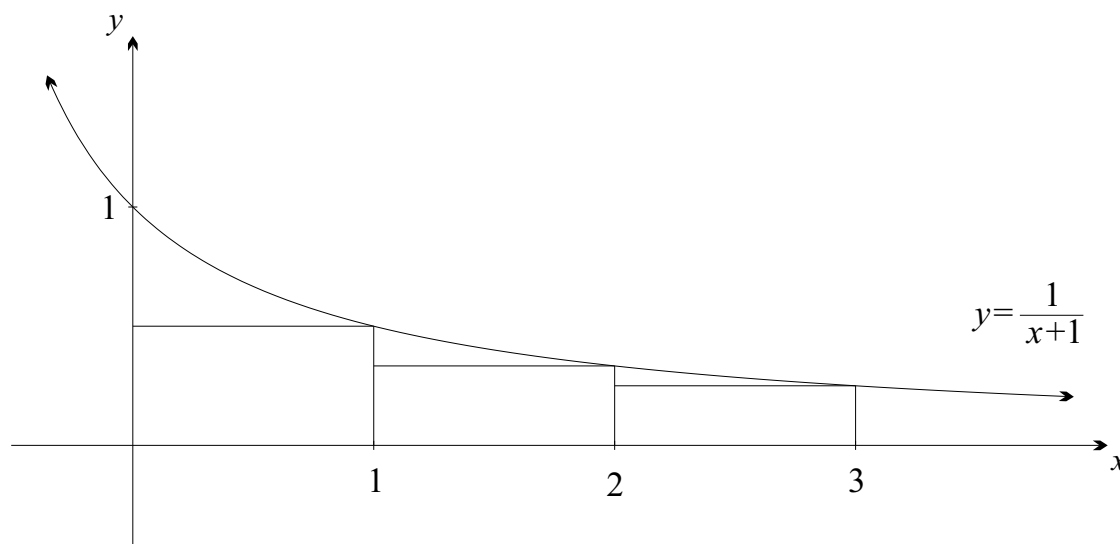
The graphs of $y = x$, $y = \frac{2}{x}$ and $x = 2$ are sketched above.

- (i) Show that A has coordinates $(\sqrt{2}, \sqrt{2})$. **1**
- (ii) Show that the area of the region bounded by the hyperbola and the two lines, shaded above, is $1 - \ln 2$. **2**
- (iii) Find the volume of the solid formed when the region is rotated around the x -axis. **3**
- (c) Solve $\sin x \tan x - 4 \sin x - \tan x + 4 = 0$, for $0^\circ \leq x \leq 360^\circ$. Give your answer correct to the nearest degree, where necessary. **3**
- (d) A geometric series has first term 35 and common ratio 2^b .
- (i) For what values of b does the series have a limiting sum? **1**
- (ii) Find the value of b for which the limiting sum is equal to 40. **2**
- (e) Determine the range of the function $y = \ln(x^2 + e)$. **1**

QUESTION SIXTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) The tangent at P on the hyperbola $xy = 4$ cuts the x and y axes at the points A and B respectively. Find all points P such that $OA : OB = 2 : 1$. **2**
- (b) Consider the quadratic $y = mx^2 + nx + mn^2$, where m and n are non-zero real numbers. For what values of m and n is the quadratic positive definite? **3**

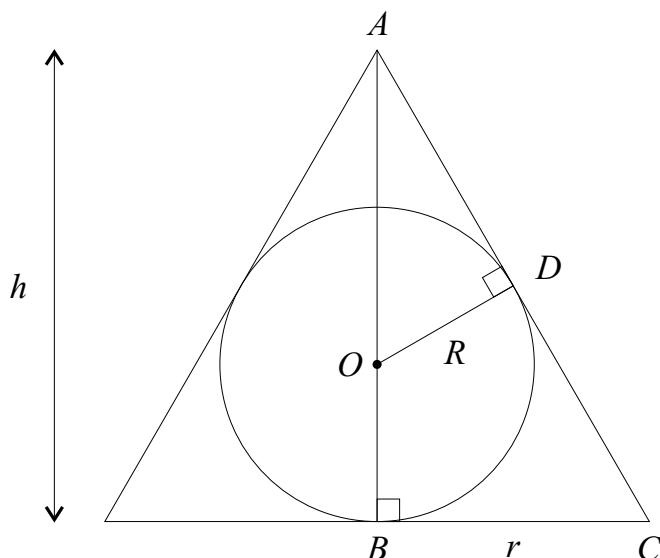
(c)



The diagram above shows part of the curve $y = \frac{1}{x+1}$ together with three rectangles of unit width.

- (i) Explain how the diagram shows that $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} < \int_0^3 \frac{1}{x+1} dx$. **1**
- (ii) The curve $y = \frac{1}{x+2}$ passes through the top left-hand corner of each of the three rectangles shown. By considering the rectangles drawn above in relation to this curve, write down a second inequality involving $\frac{1}{2} + \frac{1}{3} + \frac{1}{4}$ and a different definite integral. **1**
- (iii) By similar arguments show that $3.92 < \sum_{r=2}^{100} \frac{1}{r} < 4.61$. **2**

(d)



A cone of height h and radius r is circumscribed about a sphere with centre O and fixed radius R . The diagram above shows a cross-section of the situation through an axis of the cone AB .

- (i) Show that $r^2 = \frac{R^2 h}{h - 2R}$. 2
- (ii) Show that the volume of the cone is given by $V = \frac{\pi R^2 h^2}{3(h - 2R)}$. 1
- (iii) Find the dimensions of the cone with the least volume. You need not test that a minimum value occurs. 2
- (iv) Given the cone has minimum volume, find the ratio of the volume of the cone to the volume of the sphere. 1

_____ End of Section II _____

END OF EXAMINATION

B L A N K P A G E

NAME:

CLASS: MASTER:

SYDNEY GRAMMAR SCHOOL



2016
Annual Examination
FORM V
MATHEMATICS EXTENSION 1
Monday 5th September 2016

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

Ext 1 Solutions

mc

① $20e^{-2} = 2.706 \dots$
 $= 2.71 \quad (2 \text{ d.p.})$

A

② C

③ $3x^2 - 6x + 2 = 0$

$$\begin{aligned}x &= \frac{6 \pm \sqrt{6^2 - 4(3)(2)}}{2 \times 3} \\&= \frac{6 \pm \sqrt{12}}{6} \\&= \frac{6 \pm 2\sqrt{3}}{6} \\&= \frac{3 \pm \sqrt{3}}{3}\end{aligned}$$

D

④ $y = \frac{x^2}{x-1}$

$$\frac{dy}{dx} = \frac{(x-1) \times 2x - x^2 \times 1}{(x-1)^2}$$

$$= \frac{2x^2 - 2x - x^2}{(x-1)^2}$$

$$= \frac{x^2 - 2x}{(x-1)^2}$$

$$= \frac{x(x-2)}{(x-1)^2}$$

D

5 D

6 $7^x = 2$

$$x = \log_7 2$$

$$= \frac{\log_{10} 2}{\log_{10} 7}$$

C

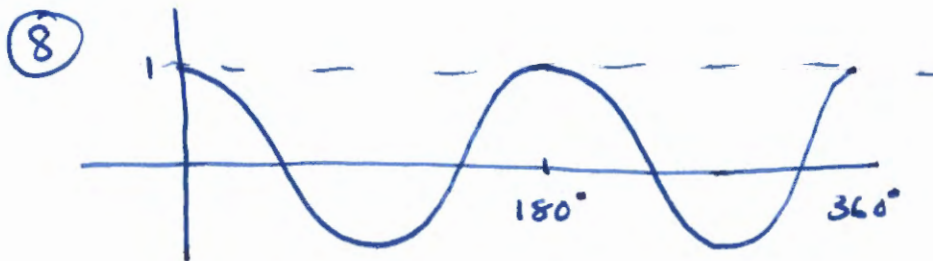
7 $\int \frac{1}{x\sqrt{x}} dx = \int \frac{1}{x \cdot x^{\frac{1}{2}}} dx$

$$= \int x^{-\frac{3}{2}} dx$$

$$= \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}}$$

$$= -\frac{2}{\sqrt{x}} + C$$

B



3 solutions

C

9 B

$$\textcircled{10} \quad \int_0^2 f(x) dx = -\frac{1}{2} \times 2 \times (5 + 2\frac{1}{2})$$

$$= -7\frac{1}{2}$$

*

$$\int_0^4 f(x) dx = -\frac{1}{2} \times 5 \times 4$$

$$= -10$$

$$\int_0^6 f(x) dx = -10 + \frac{1}{2} \times 2 \times 1$$

$$= -9$$

$$\int_0^8 f(x) dx = -10 + \frac{1}{2} \times 4 \times 1$$

$$= -8$$

A

(11)

$$(a) (2\sqrt{3}-5)(2\sqrt{3}+5) = 4 \times 3 - 5 \times 5 \\ = -13 \quad \checkmark$$

$$(b) 3x - y + 7 = 0 \\ y = 3x + 7$$

$$\text{equation: } y = 3x - 5 \quad \checkmark$$

$$(c) (x-5)(x+2) = 0 \quad \checkmark$$

$$\text{or } x^2 - (5-2)x + (5 \times -2) = 0 \\ x^2 - 3x - 10 = 0$$

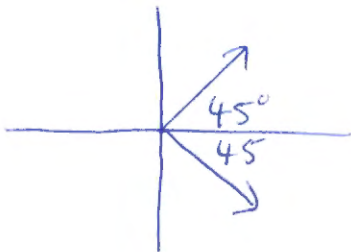
$$(d) (i) \log_e e^3 = 3 \quad \checkmark$$

$$(ii) e^{\log_e 5} = 5 \quad \checkmark$$

$$(e) x = 4 \text{ and } x = -4 \quad \checkmark$$

$$(f) \cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ \text{ or } 315^\circ \quad \checkmark$$



$$(g) m = \frac{2 - (-4)}{1 - (-2)} \\ = \frac{6}{3} \\ = 2 \quad \checkmark$$

(h) $a = 12, d = -6, n = 100$ ✓

$$S_{100} = \frac{100}{2} (2(12) + 99x - 6)$$
$$= -28500$$
 ✓

(i) $m = -\tan 30^\circ$

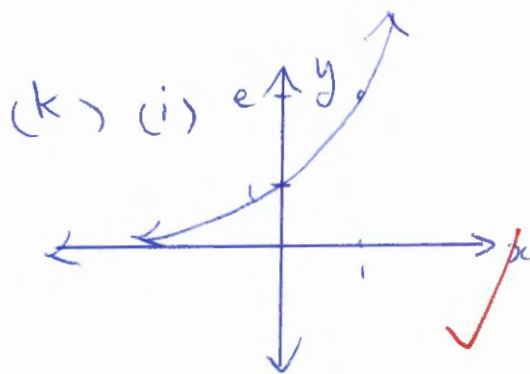
$$= -\frac{1}{\sqrt{3}}$$
 ✓

(j) $y = e^{2x+1}$

$$y' = 2e^{2x+1}$$

at $x = -1, y' = 2e^{2(-1)+1}$

$$= 2e^{-1}$$
$$= \frac{2}{e}$$
 ✓ ✓



(ii) $\int_0^1 e^x dx = [e^x]_0^1$

$$= e^1 - e^0$$
$$= e - 1$$
 ✓

12

(a) (i) $y = 3x^{-1}$
 $y' = -3x^{-2}$
 $= -\frac{3}{x^2}$ ✓

(ii) $y = \ln(5x-1)$
 $y' = \frac{5}{5x-1}$ ✓

(iii) $y = xe^x$
 $= x \times e^x + 1 \times e^x$
 $= xe^x + e^x$
 $= e^x(x+1)$ ✓

(b) (i) $\int (2x-5)^4 dx = \frac{(2x-5)^5}{5 \times 2}$
 $= \frac{1}{10} (2x-5)^5 + C$ ✓

(ii) $\int \frac{2}{x-1} dx = 2 \log_e(x-1) + C$ ✓

(iii) $\int \frac{x-1}{x^2} dx = \int \frac{1}{x} - \frac{1}{x^2} dx$ ✓
 $= \int \frac{1}{x} - x^{-2} dx$
 $= \ln x + x^{-1}$ ✓
 $= \ln x + \frac{1}{x} + C$ ✓

(c) $\frac{dy}{dx} = 3x^2 + 4x - 5$

$y = x^3 + 2x^2 - 5x + C$ ✓

Substituting $(-1, 4)$

$4 = (-1)^3 + 2(-1)^2 - 5(-1) + C$ ✓

$C = -2$

so $y = x^3 + 2x^2 - 5x - 2$ ✓

$$\begin{aligned}
 (d) \quad A &= 2 \int_0^2 (4 - x^2) dx \\
 &= 2 \left[4x - \frac{x^3}{3} \right]_0^2 \quad \checkmark \\
 &= 2 \left(4(2) - \frac{2^3}{3} - 0 \right) \\
 &= \frac{32}{3} \left(10\frac{2}{3} \right) \text{ m}^2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad x^2 + kx + 36 &= 0 \quad \checkmark \\
 \Delta &= k^2 - 4(1)(36) \\
 &= k^2 - 144 \\
 \text{If } \Delta > 0, \quad k^2 - 144 &> 0 \quad \checkmark \\
 k^2 &> 144 \\
 k < -12 \text{ or } k > 12 \quad \checkmark
 \end{aligned}$$

15

Q13

(a) $\frac{1}{2} \times m \times (m+2) \times \sin 30^\circ = 12$ ✓

$\frac{1}{2} m(m+2) \times \frac{1}{2} = 12$

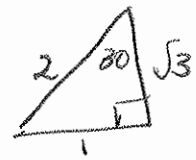
$m^2 + 2m = 48$

$m^2 + 2m - 48 = 0$ ✓

$(m+8)(m-6) = 0$

$m = -8$ or $m = 6$

as $m > 0$ choose $m = 6$ ✓



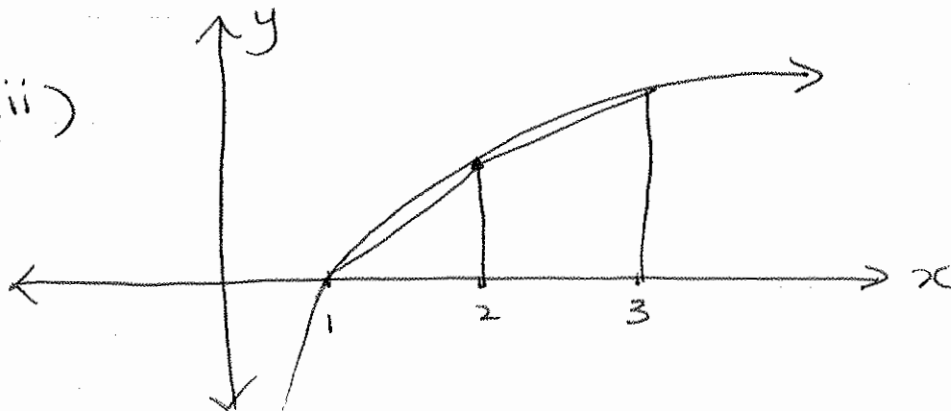
(b)

(i)

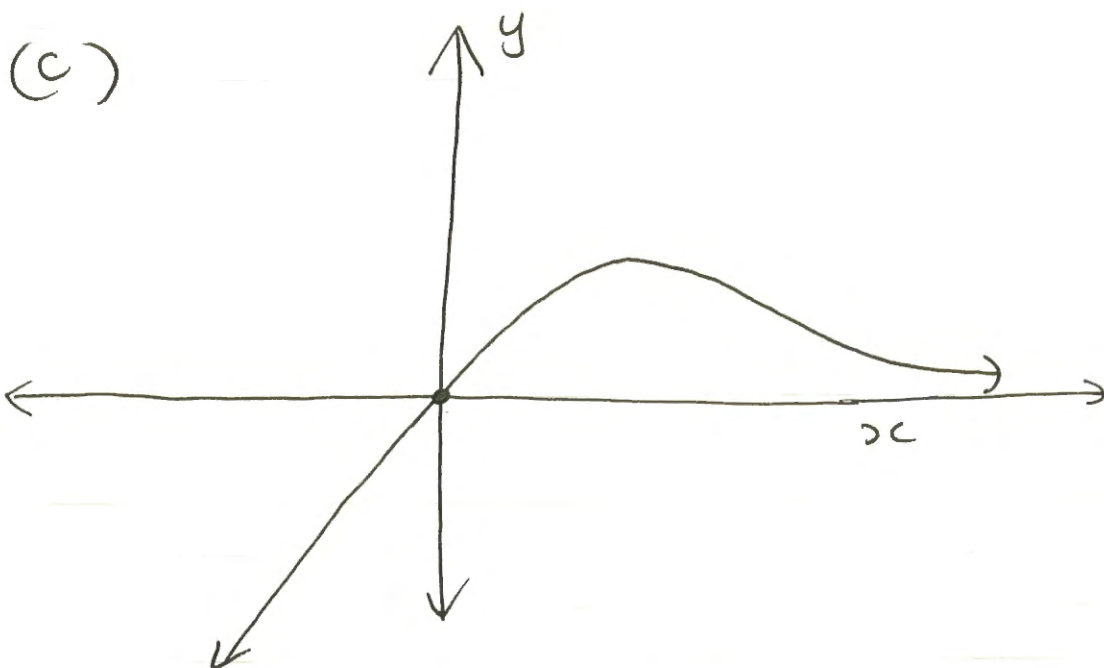
x	1	2	3
y	0	0.69	1.10

(ii) $\int_1^3 \log_e x \, dx \doteq \frac{1}{2} \times 1(0 + 0.69) + \frac{1}{2} \times 1(0.69 + 1.10)$
 $= 1.24$ ✓✓

(iii)



The triangle + trapezium underestimate the area as the curve is concave down. So the approximation will be less than the exact value. ✓



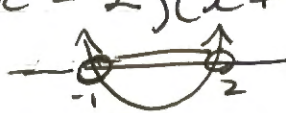
(d) (i) $f(x) = 2x^3 - 3x^2 - 12x + 4$

$f'(x) = 6x^2 - 6x - 12$

$f''(x) = 12x - 6$

character

✓ (both)

(ii) decreasing when $f'(x) < 0$
 that is, $6x^2 - 6x - 12 < 0$ ✓
 $x^2 - x - 2 < 0$
 $(x - 2)(x + 1) < 0$

 $-1 < x < 2$ ✓

(iii) concave up when $f''(x) > 0$
 that is $12x - 6 > 0$
 $12x > 6$
 $x > \frac{1}{2}$ ✓

(e) (i) $(-a, -4a)$ ✓

(ii) $(-a, 0)$ ✓

(14)

$$(a) \quad y = (x-1)^3$$

$$\sqrt[3]{y} = x-1$$

$$x = \sqrt[3]{y} + 1$$

$$A = \int_1^8 (\sqrt[3]{y} + 1) dy \quad \checkmark$$

$$= \int_1^8 (y^{\frac{1}{3}} + 1) dy$$

$$= \left[\frac{3}{4} y^{\frac{4}{3}} + y \right]_1^8 \quad \checkmark$$

$$= \left(\frac{3}{4} \times 8^{\frac{4}{3}} + 8 \right) - \left(\frac{3}{4} \times 1 + 1 \right)$$

$$= \frac{73}{4}$$

$$= 18\frac{1}{4} \quad \checkmark$$

$$(b) \quad 2x^2 - 4x + 7 = 0$$

$$(i) \quad \alpha + \beta = \frac{4}{2} \quad (ii) \quad \alpha\beta = \frac{7}{2} \quad \checkmark$$

$$= 2 \quad \checkmark$$

$$(iii) \quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{2^2 - 2 \times \frac{7}{2}}{\frac{7}{2}}$$

$$= -\frac{6}{7} \quad \checkmark$$

$$\begin{aligned}
 (c) \quad y &= (2-x)^3 \\
 y' &= 3(2-x)^2 \times -1 \\
 &= -3(2-x)^2 \\
 y'' &= -6(2-x)^1 \times -1 \\
 &= 6(2-x)
 \end{aligned}$$

$y'' = 0$ at $x = 2$, $y = 0$
possible point of inflexion at $(2, 0)$

test:

x	1	2	3
y''	6	0	-6
	↗		↘

there is a concavity change so there is a point of inflexion at $(2, 0)$

$$(d) \quad y = x^2 + \frac{16}{x}$$

$$\begin{aligned}
 &= x^2 + 16x^{-1} \\
 y' &= 2x - 16x^{-2} \\
 &= 2x - \frac{16}{x^2} \\
 &= \frac{2x^3 - 16}{x^2} \\
 &= \frac{2(x^3 - 8)}{x^2}
 \end{aligned}$$

$$\begin{aligned}
 y' = 0 \quad \text{at} \quad x^3 &= 8 \\
 x &= 2 \\
 y &= 12 \\
 y'' &= 6 \\
 &> 0
 \end{aligned}$$

there is a minimum turning point at $(2, 12)$

$$(e) \quad x^{\frac{2}{3}} + x^{\frac{1}{3}} - 6 = 0$$

let $m = x^{\frac{1}{3}}$

the equation becomes

$$m^2 + m - 6 = 0 \quad \checkmark$$

$$(m + 3)(m - 2) = 0 \quad \checkmark$$

$$m = -3 \quad \text{or} \quad m = 2 \quad \checkmark$$

$$x^{\frac{1}{3}} = -3$$

$$x = -27$$

$$x^{\frac{1}{3}} = 2$$

$$x = 8 \quad \checkmark$$

15

$$(a) \quad 3b^3 - 24 = 3(b^3 - 8)$$

$$= 3(b-2)(b^2 + 2b + 4)$$

$$(b) (i) \text{ let } \frac{z}{x} = x$$

$$x^2 = 2$$

$$x = \sqrt{2} \quad (x > 0)$$

$$y = \sqrt{2}$$

so A is the pt $(\sqrt{2}, \sqrt{2})$

$$(ii) \quad A = \int_{\sqrt{2}}^2 \left(x - \frac{z}{x}\right) dx \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{either}$$

$$= \left[\frac{x^2}{2} - 2 \ln x \right]_{\sqrt{2}}^2$$

$$= \frac{2^2}{2} - 2 \ln 2 - \left(\frac{(\sqrt{2})^2}{2} - 2 \ln(\sqrt{2}) \right)$$

$$= 2 - 2 \ln 2 - 1 + 2 \ln 2^{\frac{1}{2}}$$

$$= 1 - 2 \ln 2 + \ln 2$$

$$= 1 - \ln 2$$

$$(iii) \quad V = \pi \int_{\sqrt{2}}^2 x^2 - \left(\frac{z}{x}\right)^2 dx$$

$$= \pi \int_{\sqrt{2}}^2 x^2 - \frac{4}{x^2} dx$$

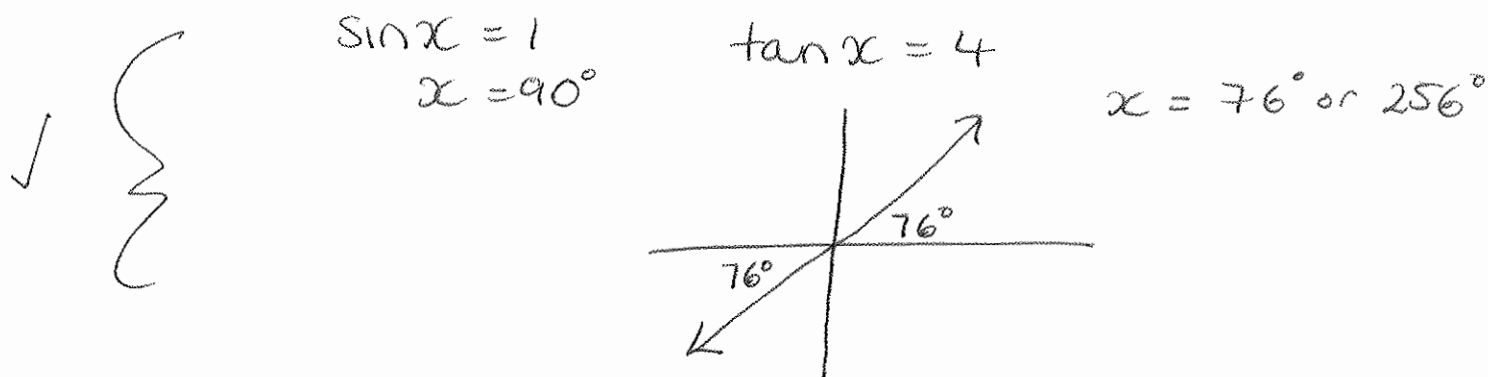
$$= \pi \left[\frac{x^3}{3} + \frac{4}{x} \right]_{\sqrt{2}}^2$$

$$= \pi \left(\frac{2^3}{3} + \frac{4}{2} - \left(\frac{(\sqrt{2})^3}{3} + \frac{4}{\sqrt{2}} \right) \right)$$

$$= \pi \left(\frac{14}{3} - \frac{2\sqrt{2}}{3} - 2\sqrt{2} \right)$$

$$= \frac{\pi}{3} (14 - 8\sqrt{2}) \quad u^3$$

$$\begin{aligned}
 (c) \quad & \sin x + \tan x - 4 \sin x - \tan x + 4 = 0 \\
 & \sin x (\tan x - 4) - 1 (\tan x - 4) = 0 \quad \checkmark \\
 & (\sin x - 1)(\tan x - 4) = 0
 \end{aligned}$$

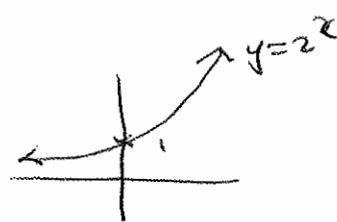


$\checkmark$ but $\tan 90^\circ$ is not defined so
 $x = 76^\circ \text{ or } 256^\circ$ are the only solutions

$$(d) \quad a = 35, \quad r = 2^x$$

$$(i) \quad -1 < 2^b < 1 \quad \checkmark$$

$$-b < 0$$



$$(ii) \quad \frac{35}{1-2^b} = 40 \quad \checkmark$$

$$1-2^b = \frac{35}{40}$$

$$2^b = \frac{1}{8}$$

$$b = -3 \quad \checkmark$$

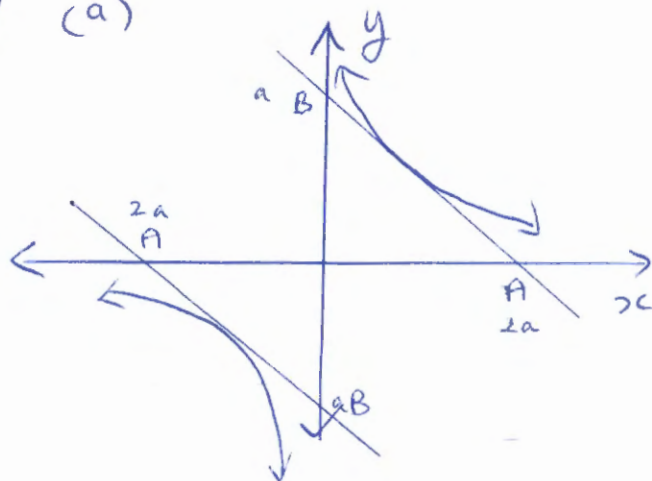
$$(e) \quad y = \ln(x^2 + e)$$

$$\text{if } x=0, \quad y = \ln e = 1$$

for any other value of x , $x^2 > 0$
 and $\therefore \ln(x^2 + e) > 1$

so range is $y \geq 1$ $\checkmark$

16 (a)



gradient of tangent
is $-\frac{1}{2}$

$$xy = 4$$

$$y = 4x^{-1}$$

$$y' = -4x^{-2}$$

$$\text{so } -\frac{4}{x^2} = -\frac{1}{2}$$

$$x^2 = 8$$

$$x = \pm 2\sqrt{2}$$

$$y = \frac{\pm 4}{2\sqrt{2}}$$

$$= \pm \sqrt{2}$$

the points are
 $(2\sqrt{2}, \sqrt{2})$ and
 $(-2\sqrt{2}, -\sqrt{2})$

(b) $y = mx^2 + nx + mn^2$

for the quadratic to be positive definite we
need $m > 0$ and $n^2 - 4(m)(mn^2) < 0$

$$n^2 - 4m^2n^2 < 0$$

$$n^2(1 - 4m^2) < 0$$

now $n^2 > 0$ for all $n \neq 0$

$$1 - 4m^2 < 0$$

$$4m^2 > 1$$

$$m^2 > \frac{1}{4}$$

$$m < -\frac{1}{2} \text{ or } m > \frac{1}{2}$$

but $m > 0$ so choose $m > \frac{1}{2}$

So the quadratic is positive definite for
all $n \neq 0, m > \frac{1}{2}$

(c) i) the area of the first rectangle is $\frac{1}{2} \times 1 = \frac{1}{2}$
 " " second " is $\frac{1}{3} \times 1 = \frac{1}{3}$
 " " third " is $\frac{1}{4} \times 1 = \frac{1}{4}$

the rectangles underestimate the area under the curve which is given by

$$\int_0^3 \frac{1}{x+1} dx$$

$$\text{so } \frac{1}{2} + \frac{1}{3} + \frac{1}{4} < \int_0^3 \frac{1}{x+1} dx$$

(ii) This time the rectangles will over-estimate the area
 so $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} > \int_0^3 \frac{1}{x+2} dx$

$$(iii) \sum_{r=2}^{100} \frac{1}{r} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{100}$$

from (i) + (ii)

$$\int_0^{99} \frac{1}{x+2} dx < \sum_{r=2}^{100} \frac{1}{r} < \int_0^{99} \frac{1}{x+1} dx$$

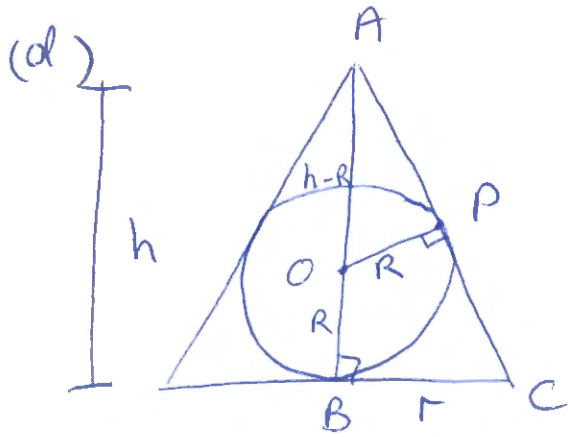
$$[\ln(x+2)]_0^{99} < \sum_{r=2}^{100} \frac{1}{r} < [\ln(x+1)]_0^{99}$$

$$\ln 101 - \ln 2 < \sum_{r=2}^{100} \frac{1}{r} < \ln 100 - \ln 1$$

$$\ln\left(\frac{101}{2}\right) < \sum_{r=2}^{100} \frac{1}{r} < \ln 100$$

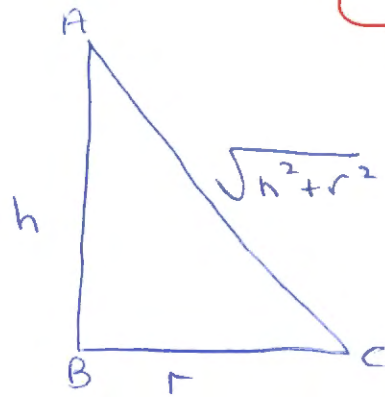
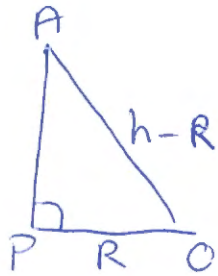
$$3.92 < \sum_{r=2}^{100} \frac{1}{r} < 4.61$$

to 2 d.p.



(i) In Δ s AOP and ABC
 $\angle APO = \angle ABC$ (given)
 $\angle A$ is common

$\Delta AOP \parallel \Delta ACB$ (angle angle similarity test)



$$\text{So } \frac{r}{R} = \frac{\sqrt{h^2 + r^2}}{h - R}$$

$$\frac{r^2}{R^2} = \frac{h^2 + r^2}{(h - R)^2}$$

$$r^2(h - R)^2 = R^2(h^2 + r^2)$$

$$r^2(h^2 - 2hR + R^2) = R^2h^2 + R^2r^2$$

$$r^2(h^2 - 2hR + R^2 - R^2) = R^2h^2$$

$$r^2 = \frac{R^2h^2}{h^2 - 2hR}$$

$$= \frac{R^2h}{h - 2R}$$

(ii) $V = \frac{\pi}{3} r^2 h$

$$= \frac{\pi}{3} \times \frac{R^2 h}{h - 2R} \times h$$

$$= \frac{\pi R^2 h^2}{3(h - 2R)}$$

$$\begin{aligned}
 \text{(iii)} \quad \frac{dV}{dh} &= \frac{1}{3} \pi \left(\frac{(h-2R) \times 2R^2 h - R^2 h^2 (1)}{(h-2R)^2} \right) \\
 &= \frac{1}{3} \pi \left(\frac{2R^2 h^2 - 4R^3 h - R^2 h^2}{(h-2R)^2} \right) \\
 &= \frac{1}{3} \pi \left(\frac{R^2 h^2 - 4R^3 h}{(h-2R)^2} \right)
 \end{aligned}$$

$$\begin{aligned}
 \frac{dV}{dh} = 0 \quad \text{at} \quad R^2 h^2 - 4R^3 h &= 0 \\
 hR^2(h - 4R) &= 0 \\
 h &= 4R, \quad h \neq 0 \\
 &\text{as } h > 2R
 \end{aligned}$$

$$\begin{aligned}
 \text{if } h = 4R, \quad r^2 &= \frac{R^2 \times 4R}{4R - 2R} \\
 &= \frac{4R^3}{2R}
 \end{aligned}$$

$$= 2R^2 \rightarrow r = \sqrt{2}R$$

least volume with $h = 4R$ and $r = \sqrt{2}R$

$$\begin{aligned}
 \text{(iv)} \quad V_C &= \frac{1}{3} \pi r^2 h \\
 &= \frac{1}{3} \pi 2R^2 \times 4R \\
 &= \frac{8\pi R^3}{3}
 \end{aligned}$$

$$V_S = \frac{4}{3} \pi R^3$$

ratio is $2:1$